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Structural groups and their mixing in borosilicate glasses of various compositions – an NMR study

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Abstract

Five binary borosilicate glasses and 54 ternary sodium borosilicate glasses covering the glass-forming regions have been examined by ¹¹B, ²³Na and ²⁹Si solid state MAS-NMR. Linear dependencies of the chemical shifts on compositional parameters were observed. In contrast to most previous structural models, the results point to a much better mixing of copolymerized SiO_{4/2}, BO_{3/2} and BO_{4/2} groups and therefore between borate and silica networks and a more uniform distribution of sodium cations in the whole composition range than hitherto assumed. Only the observed fractions of three- and four-coordinated borons and the quantitative amount of alkali to form either four-coordinated boron or non-bridging oxygen atoms cannot be described by a random model. The differences between molten and sol-gel prepared borosilicate glasses appear in a new light. © 2000 Elsevier Science B.V. All rights reserved.

1. Introduction

Borosilicate glasses do not only belong to the most widely used families of non-crystalline solids, they are also part of those glassy systems which have been investigated most frequently. But in spite of the various studies and the application of sophisticated techniques, microscopic understanding of the structure at the atomic level still has not been achieved. It was shown long time ago, especially by Bray and coworkers and Scheerer et al. [1–4] using ¹¹B wide-line NMR, that alkali borosilicate glasses behave like alkali borate glasses diluted by silica if the alkali concentration is low. BO_{3/2} and BO_{4/2} units are present and each molecular unit of alkali

oxide was thought to convert two boron atoms from three- to four-coordination. At larger alkali concentrations, Dell et al. [5] distinguished between various compositional regions characterized by the ratios $R = \text{mol\% Na}_2\text{O/mol\% B}_2\text{O}_3$ and $K = \text{mol\% SiO}_2/\text{mol\% B}_2\text{O}_3$ and postulate a highly non-statistical distribution of species. Within each region the observed fractional contributions of four-coordinated boron (N₄) and three-coordinated boron (N₃) were described by a separate set of relations as functions of R and K . The model of Dell and Bray was successfully applied to describe a number of ¹¹B wide-line NMR data.

In the meantime ¹¹B, ¹⁷O, ²³Na and ²⁹Si MAS-NMR techniques as well as Raman scattering [6] were applied to molten [6–8] and sol-gel prepared alkali borosilicate glasses [9] and such studies provided much more information than ¹¹B NMR alone. Features were observed which could not be predicted by the model referred above and it

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seemed that the distribution of structural groups appears to be more statistical than assumed.

In particular, sodium borosilicate glasses prepared by the alkoxide sol–gel method were modelled by a uniform distribution of copolymerized $\text{SiO}_{4/2}$, $\text{BO}_{3/2}$ and $\text{BO}_{4/2}$ polyhedra [9]. The simple model of homogeneously intermixed and interlinked polyhedra contrasted with the hypotheses put forward to explain the structure of molten glasses. But already first ^{29}Si MAS-NMR studies for limited molten glass compositions [6,7] showed that the silica network in alkali borate glasses must be much more intermixed with the other components than postulated in the original model. Recent ^{17}O multiple quantum magic-angle spinning NMR experiments on selected glass samples [8] demonstrated rather clearly a much better mixing between the three-coordinated borons and the silica network than in the widely accepted structural models.

It was the aim of the present study to realize whether there is really a fundamental difference between molten and sol–gel prepared borosilicate glasses as far as the randomness of the distribution of structural groupings is concerned. By expanding the number of investigated samples to the whole glass-forming region we intended furthermore to

examine earlier model studies based on a few compositions only. Therefore we prepared a large number of ternary sodium borosilicate glasses which cover the glass-forming region (Fig. 1, left-hand side). The right-hand side of the same figure represents graphically the four regions of the model of Dell et al. [5]. In addition to the ternary glasses 1–54 depicted in Fig. 1, we included also some binary B_2O_3 – SiO_2 glasses with 10, 30, 50, 70 and 90 mol% boron oxide in this study, since these systems were seldom examined by NMR.

2. Experimental

2.1. Sample preparation

Five binary borosilicate glasses and 54 ternary sodium borosilicate glasses covering the glass-forming region were prepared using reagent grade sodium carbonate, boric acid and silica powder. To shorten the relaxation times 0.2 mol% MnCO_3 was added to the batches. The glasses were melted in corundum crucibles in an inductive furnace in N_2 flow. Melting temperatures up to 1600°C were applied depending on the composition. The liquid

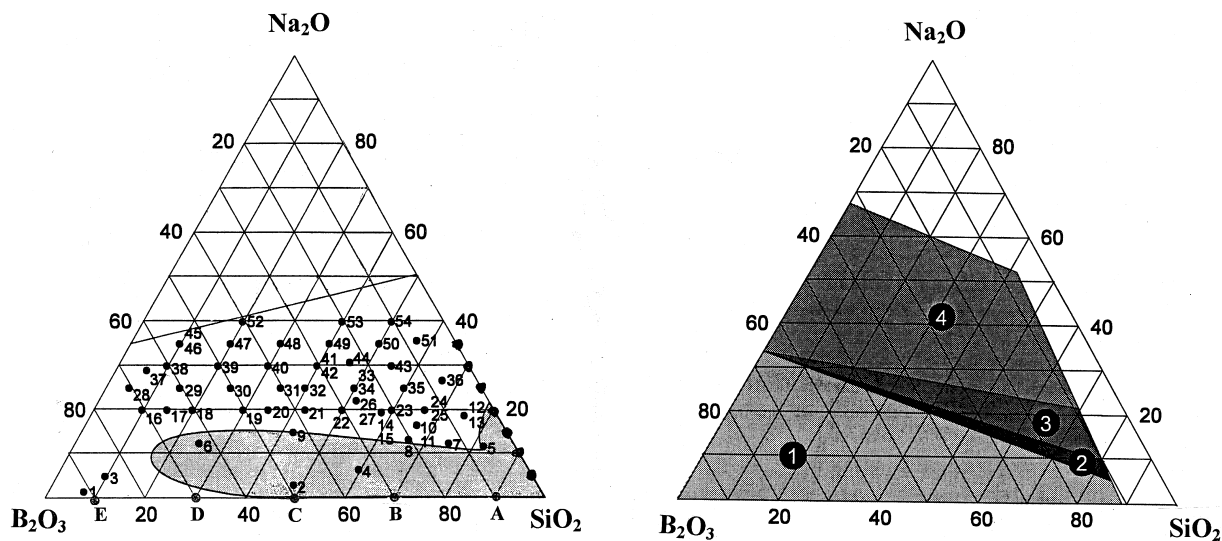


Fig. 1. Composition diagrams of the ternary system Na_2O – B_2O_3 – SiO_2 with an indication of the limit of the glass-forming region and the metastable region with a tendency of phase-separation (left). The points 1–54 and A–E refer to the samples studied within this work. The right-hand side characterizes the four regions distinguished in the model of Dell et al. 1: $R < 0.5$, 2: $0.5 \leq R \leq 0.5 + K/16$, 3: $0.5 + K/16 \leq R \leq 0.5 + K/4$, 4: $0.5 + K/4 \leq R \leq 2 + K$.

melt was maintained in the furnace until completely refined before being poured into a copper mould cooled with water.

The composition of the samples is given in Table 1, cf. also Fig. 1. Some compositions were prepared several times independently of one another for checking both the quality of the sample and the results. On the basis of detailed analyses of several glass compositions [10], the synthesis compositions were estimated to be correct within ± 1 mol%. Samples which were not completely amorphous could easily be identified and eliminated on the basis of sharp MAS-NMR peaks, especially for the ^{23}Na resonance.

The limit of the glass-forming region [11] is also indicated in Fig. 1 and as well the range where phase separation was observed at temperatures above 550°C [12]. Samples no. 2, 4, 6 and 9 lie in the immiscibility region and electron microscopy studies [10] showed indeed that these glasses are partially phase-separated.

2.2. Nuclear magnetic resonance studies

^{11}B , ^{23}Na and ^{29}Si MAS-NMR experiments were carried out on a Bruker CXP-300 FT-spectrometer using 7 mm spinning probes with a rotation rate of 5 kHz about the magic angle for ^{29}Si and 4 mm high speed spinning probes (rotation rate 17 kHz) for ^{11}B and ^{23}Na . The resonance frequencies were 59.62 MHz (^{29}Si), 79.39 MHz (^{23}Na) and 96.29 MHz (^{11}B). Typical acquisition parameters were pulse lengths of 0.6–1.8 μs , repetition times of 0.1 s (^{23}Na), 5 s (^{11}B) and 30–90 s (^{29}Si) and 100–50 000 scans. Isotropic chemical shifts are referred to $\text{BF}_3 \cdot \text{Et}_2\text{O}$, NaCl and tetramethylsilane, respectively.

One-pulse detection schemes with fairly short radio frequency pulses and long recycle delays were applied to ^{29}Si . As already observed previously [6,7] the ^{29}Si MAS-NMR spectra display no or very little resolution. They were simulated by a superposition of Gaussians and the determined chemical shifts denote centres of gravity. The ^{11}B MAS-NMR spectra were detected and interpreted as described in some detail in previous papers [9,13,14] using satellite transition spectroscopy and/or sophisticated computer programs to obtain

the Hamiltonian parameters for the $\text{BO}_{4/2}$ and $\text{BO}_{3/2}$ groups. No attempt was made to distinguish between multiple $\text{BO}_{4/2}$ sites as observed by Jäger et al. [15] since in our samples there was no clear evidence for this and it was not relevant to the objectives of this work. The focus was the quantification of three- and four-coordinated boron and the dependence of the NMR parameters on composition. The ^{23}Na MAS-NMR spectra had a similar appearance over the whole concentration range. The asymmetrically broadened resonances are indicative of second-order quadrupolar effects with wide distributions of electric field gradients. Since the shapes do not change, only the maxima of the curves were evaluated to describe the trends of the chemical shift changes.

2.3. Results

Final results are represented in terms of the compositional dependencies of the following NMR parameters (cf. Table 1):

- $\delta(\text{Si})$, the isotropic chemical shift of the centre of gravity of the ^{29}Si MAS-NMR spectra. Estimated error limits are ± 1 ppm.
- $\delta(\text{BO}_4)$ and $\delta(\text{BO}_3)$, mean values of the isotropic chemical shift of four- and three-coordinated borons. The error limits are ± 0.2 ppm and ± 0.5 ppm (up to ± 1.5 ppm if no further digit is given in Table 1), respectively. The quadrupolar parameters are not listed, as they are more or less independent of composition. For three-coordinated boron in the ternary glass systems the quadrupole coupling constant amounts to 2.6 ± 0.1 MHz and the asymmetry parameters to 0.2 ± 0.1 .
- N_4 , the fraction of four-coordinated boron atoms with an error limit of $\pm 5\%$.
- $\delta^*(\text{Na})$ the apparent chemical shift of the maxima of the ^{23}Na resonance.

3. Discussion

3.1. Binary B_2O_3 – SiO_2 glasses

The binary borosilicates A–E contain only three-coordinated boron and the shapes of both

Table 1

Composition of the investigated glass samples and experimental values of N_4 and the various isotropic chemical shifts

Sample no.	Na ₂ O (mol%)	B ₂ O ₃ (mol%)	SiO ₂ (mol%)	N ₄	$\delta(\text{BO}_4)$ (ppm)	$\delta(\text{BO}_3)$ (ppm)	$\delta(\text{Si})$ (ppm)	$\delta^*(\text{Na})$ (ppm)
A	–	10	90	–	–	13.4	–110.6	–
B	–	30	70	–	–	14.5	–110.7	–
C	–	50	50	–	–	15.8	–109.9	–
D	–	70	30	–	–	17.1	–109.2	–
E	–	90	10	–	–	17.8	–108.2	–
1	1.5	91.0	7.5	00.1	0.25	17.8	–106.3	–29.0
2	3.0	48.5	48.5	0.05	0.37	16.7	–111.0	–28.5
3	5.0	85.0	10.0	0.02	0.33	17.3	–108.0	–28.5
4	6.5	33.5	60.0	0.16	0.00	17.3	–110.0	–28.0
5	11.8	5.9	87.3	0.67	–2.03		–103.0	–28.0
6	12.5	62.5	25.0	0.24	0.00	17.7	–111.0	–28.5
7	12.5	12.5	75.0	0.85	–2.04	10	–102.2	–25.7
8	13.3	20.0	66.7	0.50	–1.54	13.7	–98.8	–30.0
9	15.0	42.5	42.5	0.39	–0.53	17.7	–112.0	–29.5
10	16.6	16.6	66.8	0.64	–1.60	12	–97.0	–29.0
11	16.6	16.6	66.8	0.77	–1.77	12		–29.0
12	18.8	6.2	75.0	0.70	–1.91		–97.8	–24.5
13	18.8	6.2	75.0	0.72	–1.81		–97.0	–24.0
14	19.5	22.3	58.2	0.42	–0.17	18.7	–98.8	–24.0
15	19.5	22.3	58.2	0.51	–1.28	15.0	–94.5	–28.0
16	20.0	70.0	10.0	0.28	0.00	17.8	–92.9	–28.5
17	20.0	65.0	15.0	0.28	–0.21	18.0	–94.0	–27.0
18	20.0	60.0	20.0	0.34	–0.25	17.1	–90.9	–27.5
19	20.0	50.0	30.0	0.28	–0.53	16.1	–93.7	–28.0
20	20.0	45.0	35.0	0.41	–0.57	17.0	–94.7	–28.0
21	20.0	37.5	42.5	0.38	–0.73	17.3	–94.9	–28.5
22	20.0	30.0	50.0	0.48	–1.25	14.5	–93.3	–28.0
23	20.0	20.0	60.0	0.59	–0.97	12.6	–93.3	–29.0
24	20.0	13.3	66.7	0.59	–1.47	12	–95.8	–28.0
25	20.0	13.3	66.7	0.67	–1.44	12	–95.0	–26.5
26	22.2	26.0	51.8	0.30	–0.90	13	–91.3	–26.0
27	22.2	26.0	51.8	0.55	–1.08	12	–93.8	–27.0
28	25.0	70.0	5.0	0.26	–0.14	18.2	–89.4	–26.0
29	25.0	60.0	15.0	0.32	–0.18	17.5	–87.3	–26.5
30	25.0	50.0	25.0	0.38	–0.29	16.8	–86.2	–26.0
31	25.0	40.0	35.0	0.45	–0.53	17.0	–89.4	–28.0
32	25.0	35.0	40.0	0.45	–0.54	17.6	–85.8	–25.0
33	25.0	25.0	50.0	0.70	–1.04	12	–89.0	–25.5
34	25.0	25.0	50.0	0.73	–0.99	12	–90.0	–
35	25.0	15.0	60.0	0.50	–1.00	11	–92.0	–23.5
36	26.6	6.7	66.7	0.68	–1.31		–94.5	–19.0
37	29.0	64.5	6.5	0.33	0.00	17.7	–87.0	–25.0
38	30.0	60.0	10.0	0.27	–0.10	17.7	–90.2	–25.0
39	30.0	50.0	20.0	0.35	–0.11	17.7	–90.6	–24.0
40	30.0	40.0	30.0	0.43	–0.34	18.6	–83.6	–23.0
41	30.0	30.0	40.0	0.44	–0.17	18.1	–86.6	–22.0
42	30.0	30.0	40.0	0.62	–0.50	16.7	–88.7	–23.0
43	30.0	15.0	55.0	0.50	–0.64	12	–90.1	–20.0
44	30.8	23.0	46.2	0.57	–0.43	13.2	–86.0	–20.0
45	35.0	55.0	10.0	0.39	0.15	18.2	–83.7	–20.0
46	35.0	55.0	10.0	0.45	0.28	17.6	–83.1	–20.0
47	35.0	45.0	20.0	0.40	0.10	18.4	–82.8	–20.0
48	35.0	35.0	30.0	0.30	0.00	19.0	–84.8	–22.0

Table 1 (Continued)

Sample no.	Na ₂ O (mol%)	B ₂ O ₃ (mol%)	SiO ₂ (mol%)	N ₄	$\delta(\text{BO}_4)$ (ppm)	$\delta(\text{BO}_3)$ (ppm)	$\delta(\text{Si})$ (ppm)	$\delta^*(\text{Na})$ (ppm)
49	35.0	25.0	40.0	0.44	−0.13	16.2	−87.1	−19.0
50	35.0	15.0	50.0	0.49	−0.24	12	−84.6	−18.0
51	35.7	7.1	57.2	0.60	−0.65		−85.4	−17.0
52	40.0	40.0	20.0	0.18	0.06	19.8	−83.5	−21.0
53	40.0	20.0	40.0	0.55	0.00	16	−79.0	−16.0
54	40.0	10.0	50.0	0.50	−0.15		−81.9	−14.0

the ^{11}B and the ^{29}Si MAS-NMR spectra do not change as the B_2O_3 content increases from 10–90 mol%. The isotropic chemical shift of ^{11}B changes linearly with the SiO_2 concentration (Fig. 2), whereas the total change of the ^{29}Si NMR is limited to 2.4 ppm only. The quadrupole coupling constant of ^{11}B amounts to 2.7 MHz for all the samples, slightly larger than that of the ternary glasses.

The relationship of Fig. 2 and the uniform decrease of $\delta(\text{BO}_3)$ with increasing SiO_2 contents indicate that the silicon and boron networks are better mixed than assumed so far [5,6]. The decrease of the chemical shift corresponds to the shielding effect of silicon in the second coordination sphere of boron already observed in selected minerals [16] and glasses [13]. The linear behaviour over an extensive range of compositions suggests that the boron and silicon polyhedra are more or less randomly distributed and connected with one another. The ‘mixing nature’ agrees with the early

IR-analysis of the spectral distribution of Si–O–Si, B–O–B and B–O–Si bands as a function of composition in binary borosilicate thin glass films [17]. The recent ^{17}O NMR study of a 40 B_2O_3 –60 SiO_2 glass by Wang and Stebbins [8] which revealed a large intensity of Si–O–B links points in the same direction.

At first glance, the ^{29}Si NMR results which do not deviate very much from those of pure silica, even at large borate concentrations, seem to contradict such an interpretation. They may be explained, however, by bond strength calculations which showed that Si–O–Si and Si–O–B(III) bonds should have similar ^{29}Si chemical shifts [18,19]. A sol–gel prepared 5 B_2O_3 –95 SiO_2 glass of a previous study displayed a less statistical nature; only 66% of the $\text{BO}_{3/2}$ units were found to be connected with silica tetrahedra whereas the rest formed a borate network [9].

3.2. Sodium borosilicate glasses

Inspection of the results of Table 1 shows certain trends as far as the compositional dependencies of the parameters N_4 , $\delta(\text{BO}_4)$, $\delta(\text{BO}_3)$, $\delta(\text{Si})$ and $\delta^*(\text{Na})$ are concerned, but it does not point to distinct regions with highly non-statistical species distributions such as those of Fig. 1, right-hand side.

N_4 tends to increase at increasing SiO_2 content and to decrease with the borate fraction; apart from very low sodium oxide concentrations there is no clear dependence on the Na_2O contents. The larger amount of N_4 data does not really contradict previous results which were used for the development of specific models [5], but it does not suggest necessarily such models. In connection

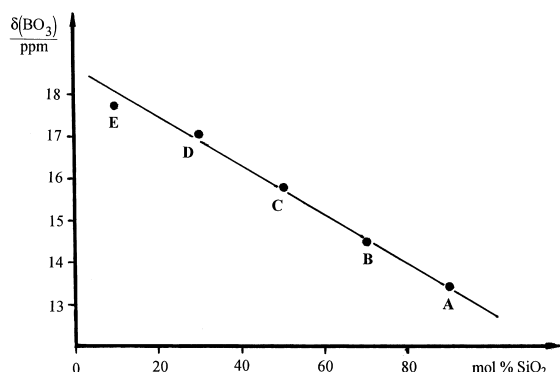


Fig. 2. Plot of the observed isotropic chemical shift $\delta(\text{BO}_3)$ against the silica contents of binary borosilicate glasses (cf. Table 1).

with the chemical shift data the results point to a more statistical distribution of structural units. N_4 measurements are more widely spread than those of the chemical shifts depending also on the sample preparation.

$\delta(\text{BO}_3)$ varies between 10 and 20 ppm and similar to the binary borosilicates it decreases with increasing SiO_2 content. There is no clear dependence upon the Na_2O fraction. $\delta(\text{BO}_4)$, on the other hand, tends to increase with the amount of Na_2O and to decrease with the silica fraction. The total range of values lies between -2 and $+0.7$ ppm. No special behaviour is observed for the distinguished regions of the model of Dell et al. ([5] and Fig. 1), but samples nos. 1, 2, 3, 4, 6 located in the metastable region of the compositional diagram of Fig. 1 do not fit in with these systematics.

$\delta(\text{Si})$ increases clearly from the value of -110 ppm for a pure silica network to about -80 ppm upon introducing sodium ions. There is no clear dependence on the B_2O_3 fraction; $\delta^*(\text{Na})$ varies between -30 and -14 ppm with a certain tendency to increase with the Na_2O content.

The large amount of data cannot be explained in terms of existing models. Considering the model of Dell et al. [5], for the compositional region 1 (cf. Fig. 1) $\delta(\text{Si})$ should not be far from -110 ppm and $\delta(\text{BO}_4)$, $\delta(\text{BO}_3)$, $\delta(\text{Na})$ and N_4 were expected to behave similarly as in binary sodium borate glasses. Inspection of Table 1 shows that this is not the case for the majority of our 16 samples belonging to this region.

The four samples located in region 2 of the model reveal smaller N_4 values than expected and considering the various results there is no indication of reedmergnerite groups which were previously invoked.

In region 3 we studied 16 samples, in region 4 15 samples. In general, N_4 appears smaller than predicted by the model and the exclusive formation of non-bridging oxygen on the silica tetrahedra by the additional sodium oxide in 3, as well as the formation of pyroborate units and silica tetrahedra with two non-bridging oxygens per Si atom in 4 cannot be verified by the NMR parameters.

A satisfactory description of the results is more possible by a uniform distribution of copolymer-

ized $\text{SiO}_{4/2}$, $\text{BO}_{3/2}$ and $\text{BO}_{4/2}$ polyhedra as previously applied to sodium borosilicate glasses prepared by the alkoxide-gel method [9]. The structure of the molten glasses is therefore not that much different as originally anticipated. The statistical model used in [9] calculates $\delta(\text{BO}_4)$ from the composition by assuming that each $\text{SiO}_{4/2}$ and $\text{BO}_{4/2}$ tetrahedron linked to a $\text{BO}_{4/2}$ group reduces the shift by 0.5 ppm, as well as each $\text{BO}_{3/2}$ triangle in the second coordination sphere. $\delta(\text{Si})$ is obtained from considering the superposition of the various Q^n groups and the percentage with which the group occurs. The Na_2O content enters indirectly since it influences both the concentration $[Q^n]$ and the individual shift $\delta(Q^n)$. Using this model the present data can satisfactorily be described for glasses with medium-sized sodium oxide concentrations. An improved statistical model, which accounts in a more detailed way for the role of sodium oxide and which describes the compositional dependence of the multitude of results rather well, will be presented in the next section.

3.3. A simple statistical model

Most parameters can be modelled by linear dependencies on the concentration of the various structural units, i.e. $\text{BO}_{3/2}$, $\text{BO}_{4/2}$, $\text{SiO}_{4/2}$ and Na^+ . We assume homogeneously intermixed and interlinked polyhedra, but no clear relation exists as far as the formation of four-coordinated boron is concerned. The relative number of $\text{BO}_{4/2}$ units at a given composition depends also on the thermal history of the glass and enters as an experimental parameter in the description.

We consider first the observed chemical shift of boron nuclei in tetrahedral groups, $\delta(\text{BO}_4)$. Silicon linked to $\text{BO}_{4/2}$ groups via bridging oxygens increases the shielding and causes a negative chemical shift. Introduction of sodium ions, on the other hand, leads to more positive shifts. The large majority of the measured $\delta(\text{BO}_4)$ values can be described by a linear combination of both effects

$$\delta(\text{BO}_4)/\text{ppm} = -3.4x_{\text{Si}} + 2.8x_{\text{Na}} - 0.62, \quad (1)$$

x_{Si} and x_{Na} are the mole fractions of silica and sodium oxide, respectively, in the glasses

$$x_{\text{Si}} = \frac{\text{mol}\% \text{ SiO}_2}{\text{mol}\% \text{ SiO}_2 + 2\text{mol}\% \text{ B}_2\text{O}_3 + 2\text{mol}\% \text{ Na}_2\text{O}},$$

$$x_{\text{Na}} = \frac{2\text{mol}\% \text{ Na}_2\text{O}}{\text{mol}\% \text{ SiO}_2 + 2\text{mol}\% \text{ B}_2\text{O}_3 + 2\text{mol}\% \text{ Na}_2\text{O}}.$$

Fig. 3 shows a plot of $\delta(\text{BO}_4)$ calculated with Eq. (1) inserting the respective compositions (cf. Table 1) plotted against the experimental values. The agreement looks quite well, with the exception of glasses with very small sodium concentrations and those lying in the immiscibility region (cf. Fig. 1). Glass no. 14 with a limiting composition was prepared a second time (glass no. 15) and the new sample fits the relationship very well. All the deviations mirror an apparent Na_2O concentration which is smaller than that corresponding to the actual composition.

The validity of the linear superposition for the whole range of compositions emphasizes the statistical character of the intermixing of the various polyhedra in the glass. The limiting value of $\delta(\text{BO}_4) = -0.62$ ppm characterizes an environment of few existing $\text{BO}_{4/2}$ groups which consists essentially of $\text{BO}_{3/2}$ units. Replacement of $\text{BO}_{3/2}$ by $\text{SiO}_{4/2}$ in the second coordination sphere of boron (IV) makes the shift more negative because of the increased shielding. The role of sodium ions is only observable as the overall effect: $\text{BO}_{4/2}^-$,

$\text{BO}_{2/2}\text{O}^-$ and $\text{SiOO}_{3/2}^-$ may be formed in the second coordination spheres and the total shift is increased.

A simple linear relationship was also found for the observed $\delta(\text{Si})$ values

$$\delta(\text{Si})/\text{ppm} = 86.2N_4x_{\text{B}} + 57.9x_{\text{Na}}^{\text{rest}} - 11x_{\text{Si}} - 111 \quad (2)$$

with

$$x_{\text{B}} = \frac{2\text{mol}\% \text{ B}_2\text{O}_3}{\text{mol}\% \text{ SiO}_2 + 2\text{mol}\% \text{ B}_2\text{O}_3 + 2\text{mol}\% \text{ Na}_2\text{O}}.$$

This is shown in Fig. 4 and the results of previous measurements [7] fit this line perfectly as well (not shown). Exceptions are the phase-separated glasses nos. 6 and 9 which mirror a complete silica environment in the second coordination sphere and this in spite of the presence of more than 10 mol% Na_2O . In Eq. (2) N_4x_{B} means the mole fraction of four-coordinated boron and

$$x_{\text{Na}}^{\text{rest}} = x_{\text{Na}} - N_4x_{\text{B}}$$

is that part of the mole fraction of sodium oxide which is not consumed to produce four-coordinated boron, but available to form SiO_4 units with non-bridging oxygen atoms (Q^3 and Q^2 units).

As already stated in connection with the binary borosilicate glasses, $\text{BO}_{3/2}$ groups in the second coordination sphere of silicon do not influence the chemical shift. Therefore, the chemical shift $\delta(\text{Si})$ is increased (from the negative value of

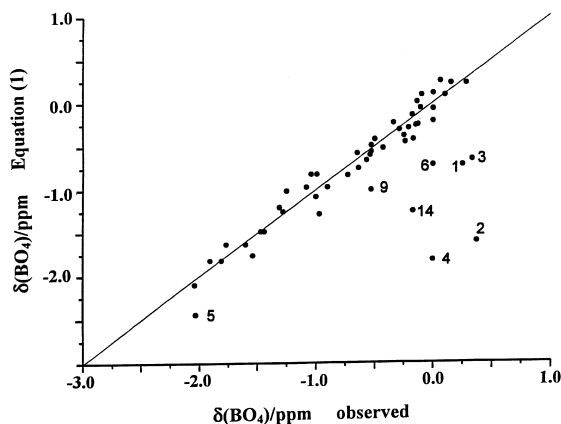


Fig. 3. $\delta(\text{BO}_4)$ values as calculated for all the samples after Eq. (1) plotted versus the experimental $\delta(\text{BO}_4)$ data of Table 1. To allow a clear view the numbers of the glass samples were omitted. Only the numbers of those samples are indicated which do not follow the relation.

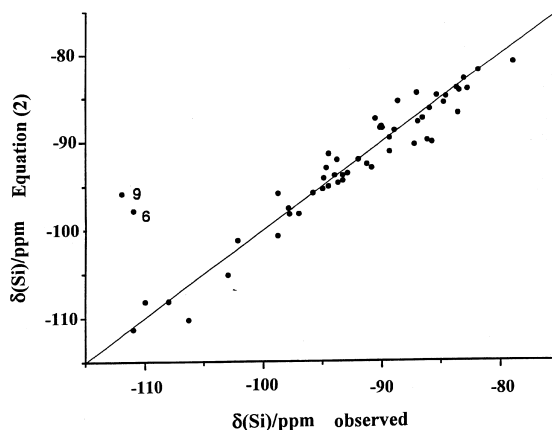


Fig. 4. Same as Fig. 3, but for $\delta(\text{Si})$ and Eq. (2).

pure silica) only by the fraction of four-coordinated boron and by the formation of silica Q^n units with $n < 4$. The validity of Eq. (2) for the whole range of compositions, Fig. 4, emphasizes again the statistical character of intermixing. In contrast to Eq. (1), however, Eq. (2) does not only contain compositional parameters, but in addition the experimentally obtained fraction of $BO_{4/2}$ units. Different from the chemical shifts, N_4 cannot be obtained in a simple way just from the composition; it depends also on the thermal history of the glass; different N_4 values were observed in different samples of identical compositions.

The apparent chemical shift of the ^{23}Na resonance can be represented rather well by a linear dependence on the mole fraction of sodium oxide

$$\delta^*(\text{Na})/\text{ppm} = 40x_{\text{Na}} - 37 \quad (3)$$

as shown in Fig. 5. Exceptions are again glasses in the immiscibility region distinguished by clustering. The behaviour of all the other glasses indicates a rather uniform distribution of cations in the network.

$\delta(\text{BO}_3)$ finally can only roughly be modelled analogously to the binary glasses (cf. Fig. 2) as far as the dependence on the silica contents is concerned, Fig. 6. Both Figs. 2 and 6 point to a mixing of three-coordinated boron with the silica network.

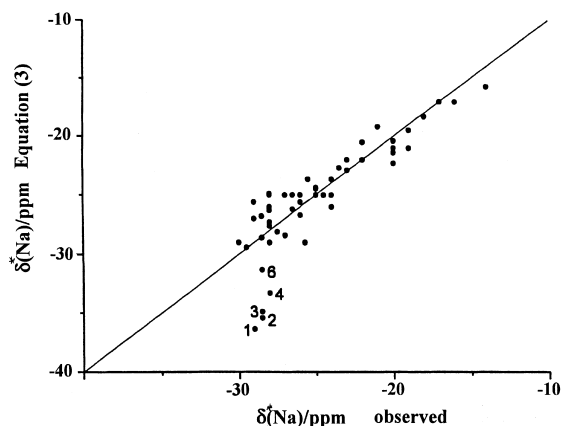


Fig. 5. Same as Fig. 3, but for $\delta^*(\text{Na})$ and Eq. (3).

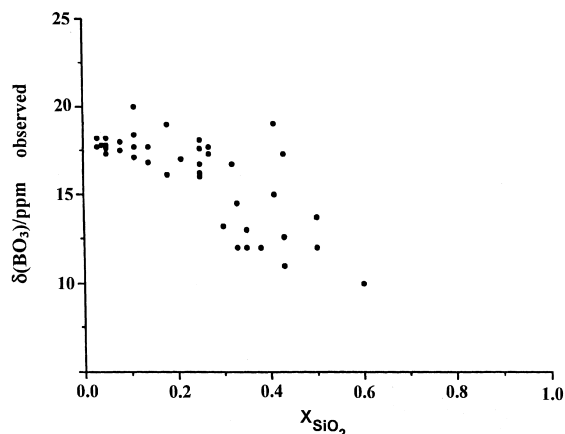


Fig. 6. Plot of $\delta(\text{BO}_3)$ values against the mole fraction of silica. The sample numbers were omitted again.

4. Conclusions

Our measurements over an extended range of glass compositions and the observed linear dependencies of chemical shift values reveal that the borate and silica networks as well as the cations are statistically mixed to a greater extent than has been assumed in most models of borosilicate glasses. There is no indication of distinct compositional regimes with preferred environments of specific groups, but a large distribution of local environments for the whole concentration range. Exceptions are only glasses within the known immiscibility region which are demonstrably phase-separated. The best model at present seems to be a random mixing and copolymerization of $\text{SiO}_{4/2}$, $\text{BO}_{2/2}$ and $\text{BO}_{4/2}$ polyhedra and a homogeneous distribution of cations at any given concentration. But the relative number of $\text{BO}_{3/2}$ and $\text{BO}_{4/2}$ units as well as the fractions of sodium consumed to form either four-coordinated boron or non-bridging oxygen atoms can only be determined experimentally and not yet modelled.

A further consequence of these findings is the difference between molten and sol-gel prepared borosilicate glasses discussed in [9]. The random character should no longer be viewed as a special feature of the gel-prepared glasses, as initially thought. The chemical shifts observed in those

glasses were described by a model for the second coordination sphere of boron and silicon nuclei which takes account the chemical shift increments of the various groups corresponding to their share at a given composition. Sodium ions were only indirectly considered by their effect of converting $\text{BO}_{3/2}$ and $\text{SiO}_{4/2}$ units. Such a model is not bad in describing the molten glasses, but there are systematic discrepancies: the calculated $\delta(\text{BO}_4)$ appears too large at small Na_2O concentrations and too small at large ones. $\delta(\text{Si})$ appears too large (less negative) for glasses with small silica contents. Alkoxide-gel prepared glasses, on the other hand, can be described better by the increment model than by the linear combinations of Eqs. (1) and (2).

These observations point to structural differences which are mainly connected with the influence of Na^+ ions rather than with the statistical mixing of all structural units. In sol–gel glasses containing only small fractions of silica the effect of the cations appears to be more pronounced than in molten glasses. Differences in the number of effective structural units exist furthermore if the Na^+ concentrations are either rather small or very large. It seems that the cations are more homogeneously distributed in molten glasses and within the available space in gel glasses as well. Furthermore, a special structural role of residual hydrogen atoms may also play a role.

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